CHEM 427/527 · SESSION 7 · THURSDAY, OCTOBER 22, 2026

Visualization for chemical measurements

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Drafted with assistance by Claude Code. Reviewed and revised by Matt.

LEARNING OBJECTIVES

By the end of today you can

1. Read a hyperspectral cube one band at a time, and find the bands that carry no signal and the pixels outside the image.
2. Make a band-ratio map with a perceptually uniform colormap, percentile color limits, and a labeled colorbar. Show in grayscale the boundaries a rainbow colormap invents.
3. Plot each polymer's mean spectrum beside the map, and choose which pixels go into each mean when lamp glare clips some of them.
4. Plan a figure in words before any code, and change the plan where it would hide the difference between PET and PVC.

FIRST, EVERYONE PULLS

Get session 7 with git pull

```
git pull
```

S07-visualization appears, with

- **data/apr22/**: 81 band images and a mask
- **data/apr22_spectra.csv**: the survey's mean spectra
- **data/batch/**: 76 band images and four masks
- **data/band_pairs.csv**: every band pair

Session 6 is due today

- Upload it by 1:30 pm.
- Restart and Run All before you upload.

Nothing to download today: the survey crop, the test batch, and every band pair are in the repository.

FROM A FOLDER OF FILES TO ONE IMAGE

What is the spectrum of one pixel?

Two of the 81 files in data/apr22/

```
sequence_1_SWIR_fl01500_1659_0900.png
…
sequence_1_SWIR_fl01500_1659_1700.png
```

The name ends in the wavelength in nm, so name order is wavelength order.

A band

One grayscale image for one narrow band of wavelengths.

The cube

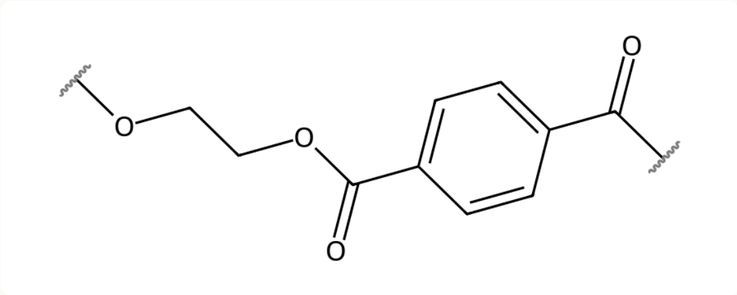
The bands stacked in wavelength order: two axes of position, one of wavelength.

A pixel's spectrum

The same pixel read from every band in turn.

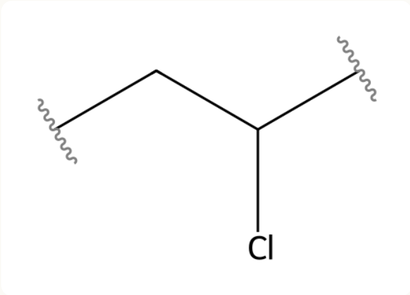
SECTION 0 OF THE NOTES

Two clear plastics to tell apart



PET

Poly(ethylene terephthalate), a polyester. Drink bottles and food trays; resin code 1.



PVC

Poly(vinyl chloride), a chlorine on every second carbon. Pipes, window frames, and blister packs; resin code 3.

Above 250 °C, where PET is melted for reuse, PVC releases hydrogen chloride, which breaks PET's ester bonds and yellows the resin.

THE DECISION

Which flakes go to the PET bale?

The operator

Reads a false-color map from a SWIR camera over a belt of mixed flakes, and sends each flake to a bin.

PVC in the PET bale

Spoils the melt, and the reclaimer rejects the whole bale.

PET in another bin

Goes to landfill.

Real: data/apr22/

A drone survey of plastic objects laid on grass, flown in April 2022 by Balsi, Moroni, and Bouchelaghem.

Simulated: data/batch/

A test batch on a belt, built from that survey's measured spectra, with brightness, thickness, and glare modeled.

The plant's own rule: a flake goes to the bin its spectrum matches.

LIVE, IN A SCRATCH NOTEBOOK: ONE RAMP, TWO COLORMAPS

Which ramp has edges in it?

ramp = np.tile(np.linspace(0, 1, 256), (20, 1)), a smooth rise from 0 to 1. In each card, the top strip is the colormap and the bottom strip is its grayscale print, from as_gray in section 1.

jet

```
imshow(ramp, cmap="jet")
```



The print rises to near white in the middle and falls to dark at both ends.

viridis

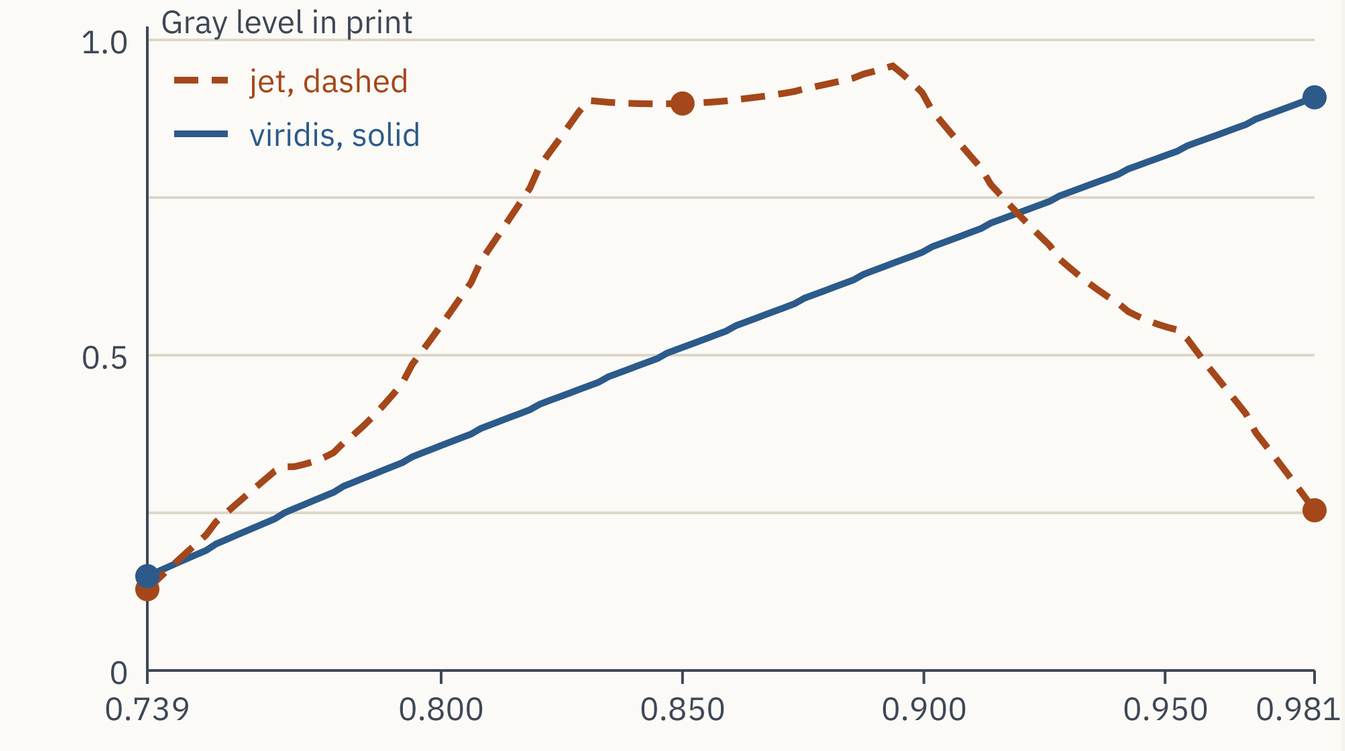
```
imshow(ramp, cmap="viridis")
```



The print rises evenly from dark to light.

SECTION 1, THE WORKED EXAMPLE ON THE APRIL 2022 CROP

Where jet's gray turns over



Color limits at the crop's 1st and 99th percentiles, **0.739** and **0.981**.

Ratio	jet	viridis
0.739	0.13	0.15
0.850	0.90	0.51
0.981	0.25	0.91

Jet prints a bright ring where the ratio passes 0.85, and prints the PVC core and the brightest grass alike.

ASSIGNMENTS

One band pair each

#	Pair	Bands, nm
1	pair_01	1540 / 1190
2	pair_02	1210 / 1230
3	pair_03	1300 / 1200
4	pair_04	1220 / 1060
5	pair_05	1530 / 1160
6	pair_06	1050 / 1170
7	pair_07	1070 / 1170
8	pair_08	1170 / 1010
9	pair_09	1090 / 1210

#	Pair	Bands, nm
10	pair_10	1600 / 1180
11	pair_11	1220 / 1250
12	pair_12	1330 / 1160
13	pair_13	1560 / 1170
14	pair_14	1220 / 1080
15	pair_15	1240 / 1210
16	pair_16	1170 / 1570
17	pair_17	1190 / 1330
18	pair_18	1180 / 1550

#	Pair	Bands, nm
19	pair_19	1230 / 1180
20	pair_20	1290 / 1180
21	pair_21	1220 / 1210
22	pair_22	1210 / 1560
23	pair_23	1210 / 980
24	pair_24	1160 / 1630
25	pair_25	1590 / 1160
26	pair_26	1170 / 1300

In Parameters, set **NUMERATOR_NM** to the first band and **DENOMINATOR_NM** to the second: pair_01 sets 1540 and 1190.

AI PRACTICE 4 OF 8: EXPLORE AND PLAN BEFORE CODE

The plan comes before the figure

Live, in Claude Code, with the scratch notebook open

/plan A figure of ramp for a reader who will print it in grayscale. Say in words which colormap and limits you would use and why, and how the colorbar is labeled. No code until I approve.

No /plan in the menu? Click the mode indicator under the prompt box and choose **Plan**.

Read in the plan, before any code

- **The colormap.** Does its gray print keep the order of the values?
- **The color limits.** Minimum and maximum, or percentiles?
- **The colorbar label.** The quantity and both bands?

In a plan, a colormap costs a comment to change. Before TASK 2, the same move on your pair.

THE JUDGMENT CELL, SECTION 4

Your display, and one entry

(a) Your answer

- Does your figure let the operator tell PET from PVC?
- Which treatment would you build the display on?
- Your medians and the PVC minus PET difference under both treatments

(b) One judgment entry

- What the assistant suggested
- What you did: changed, rejected, or verified
- Why, from a value your notebook printed

Scored 0 to 2. The plan before TASK 2 is the source; the range against the percentiles, or a gray level, is the evidence to cite.

SECTION 1 IS READ AND RUN. SECTION 2 IS YOURS.

Go, then submit

Submit

Restart, Run All, save, and upload activity.ipynb by **1:30 pm Tuesday, October 27**.

Every band pair is in the repository, so the graders' run reads the same data yours did.

Scoring, 0 to 3

1 if it runs after a restart, **0** if not.

0 to 2 for the judgment cell.

Help

Ask Matt, Lucas, or Chris in the room.

The third hour is an office hour.